Victor *et al. BMC Pregnancy and Childbirth* (2025) 25:1339
https://doi.org/10.1186/s12884-025-08536-6

BMC Pregnancy and Childbirth

RESEARCH Open Access

National and regional Temporal trends and forecasting of preterm birth in brazil: evidence from National birth data (2014–2023) with projections to 2030

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Abstract

Background Preterm birth (PTB) remains one of the leading causes of neonatal morbidity and mortality globally. In Brazil, regional disparities and socioeconomic inequalities significantly affect maternal and perinatal outcomes. This study aims to describe national and regional trends in preterm birth in Brazil from 2014 to 2023, examine sociodemographic disparities, and forecast future rates through 2030 using time-series analysis.

Methods We conducted a population-based ecological analysis using data from the Brazilian Live Birth Information System (SINASC) from 2014 to 2023. PTB was defined as delivery before 37 completed weeks of gestation. Descriptive statistics were employed to explore maternal, obstetric, and neonatal characteristics. Temporal trends were assessed using Prais-Winsten regression to estimate Annual Percentage Change (APC) and Average Annual Percentage Change (AAPC) at a significance level of 5%. A national monthly time series was constructed using Forecasting models, including Seasonal Autoregressive Integrated Moving Average (SARIMA) and Generalised Autoregressive Conditional Heteroskedasticity (GARCH).

Results The analysis included over 25.5 million live births. The national prevalence of PTB increased from 11.3% in 2014 to 11.9% in 2023, with an AAPC of 0.28% (95% CI: 0.43 to 1.00; $p < 0.001$). Higher rates were consistently observed among those without a formal education (15.1%). Regional disparities were evident, with the highest rates among mothers aged ≥ 35 years (13.6%) and those aged < 20 years (12.8%), as well as among those in the North (12.2%) and Northeast (11.5%) regions. SARIMA modelling indicated a continued upward trajectory through 2030. In contrast, more educated women (≥ 12 years) are expected to maintain lower and stable rates (< 13%).

Conclusions PTB rates in Brazil have shown a rising trend over the past decade, with marked regional and social inequalities. Forecasts indicate that these disparities may increase by 2030 unless effective, evidence-based, and context-specific interventions are implemented. These findings highlight the urgent need for targeted public health

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policies and prenatal care strategies aimed at reducing preventable PTB and promoting maternal and neonatal equity across the country.

Keywords Preterm birth, Temporal trends, Forecasting, SARIMA, GARCH and SINASC

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Introduction

Preterm birth (PTB), defined as delivery before 37 weeks of gestation, remains a significant global public health challenge. It is the leading cause of neonatal mortality and morbidity worldwide, contributing significantly to acute complications, long-term neurodevelopmental impairments, and chronic conditions such as cardiometabolic diseases and obesity [1–3]. In 2020, an estimated 15 million preterm live births were recorded globally, representing a prevalence of approximately 9% [4]. While high-income countries have established robust surveillance systems, reliable data remain limited in many low- and middle-income countries (LMICs), despite over 80% of global births occurring in health facilities [2, 4]. In Southern Asia, the PTB rate reached 13.2%, while in Latin America, it ranged from 8% to 12% [2, 4, 5]. Addressing PTB is essential to achieving Sustainable Development Goal 3.2, which aims to reduce neonatal mortality to 12 deaths per 1,000 live births by 2030 [4].

In Brazil, despite advances in maternal and child health, the prevalence of PTB remains stubbornly high at around 11% [6]. Brazil is among the top ten countries with the highest absolute number of PTBs [1]. Regional and sociodemographic disparities are well documented: the North and Northeast regions report consistently higher rates, and adolescents, women aged 35 and above, Black and Indigenous mothers, and those with low schooling or insufficient prenatal care face greater risks [6–8]. These inequities mirror patterns observed in high-income countries, where racial and social determinants significantly influence PTB outcomes [9–11]. The clinical and economic impact of PTB is significant, including neonatal intensive care unit (NICU) admissions, extended hospital stays, higher morbidity rates, and long-term follow-up needs. These factors place substantial pressure on the Brazilian Unified Health System (SUS). Additionally, the distribution of public high-complexity NICUs is uneven across the country, making it essential to understand the determinants and trends associated with PTB in Brazil [6, 12].

PTB is now widely recognised as a multifactorial syndrome rather than a single disease entity. Its aetiology includes infections, inflammation, uteroplacental insufficiency, uterine overdistension, stress, and immunologic processes[3, 13, 14]. A definitive cause is often elusive, and multiple pathways may coexist. In high-income countries like the U.S., a growing share of preterm deliveries is medically indicated and associated with declines in perinatal mortality [15], underscoring the importance of distinguishing between spontaneous and provider-initiated PTB when analysing trends and formulating policies.

The COVID-19 pandemic represented an additional turning point for maternal and perinatal health, with potential implications for PTB trends. Several studies have reported that disruptions in prenatal care, changes in obstetric practices, and indirect social effects during the pandemic period have influenced PTB patterns in Brazil and worldwide [6, 16]. However, evidence remains inconsistent, with some regions reporting decreases in medically indicated PTB, while others observed increases among socially vulnerable groups. Thus, incorporating the pandemic period is crucial to understanding shifts in obstetric care and health inequalities. Despite recent findings, the long-term impact of the pandemic on PTB trends across Brazil remains poorly understood.

Although several studies have investigated PTB prevalence in Brazil using data from the Live Birth Information System (SINASC), most analyses were restricted to specific regions or limited time frames, often without accounting for post-pandemic changes or future projections [6, 7, 16]. Moreover, few studies have applied time-series forecasting methods capable of predicting future trends and quantifying the persistence of social inequalities [17]. In the context of an evolving epidemiological landscape and increasing data availability, monitoring historical trends alone is no longer sufficient. Forecasting future PTB scenarios can provide policymakers with valuable insights, enabling better anticipation of resource needs and prevention priorities, particularly in populations historically exposed to structural inequalities.

Ecological time-series studies, such as the present one, are particularly suitable for this type of analysis [18]. This approach allows for the identification of temporal patterns, assessment of long-term fluctuations, and comparison between states and regions an essential aspect in a country characterized by marked territorial and social heterogeneity such as Brazil. Therefore, this study aimed to describe national and regional trends in preterm birth in Brazil from 2014 to 2023, examine sociodemographic disparities, and forecast PTB rates through 2030 using an ecological time-series design based on national birth registry data.

Methods

This population-based ecological study analysed nationwide data on live births in Brazil from 2014 to 2023 to estimate the prevalence, temporal trends, and forecast of PTB. The unit of analysis was the Brazilian state, with data aggregated by state and year. This period was selected due to the availability of complete and standardized SINASC data following system improvements implemented in 2014 and to exclude the incomplete year of 2024, ensuring data comparability and temporal consistency.

Data were obtained from the Live Birth Information System (SINASC), established in 1990 by the Department of Informatics of the Unified Health System (DATASUS), Ministry of Health, Brazil. SINASC aims to collect and consolidate information on all live births occurring across the country through mandatory reporting by healthcare professionals or traditional birth attendants via the declaration of live birth [19]. After completion of each DNV form, the data are processed by the Municipal Health Departments, then consolidated at the state and national levels. SINASC is widely recognized for its high coverage and data quality and is frequently used in epidemiological research for analyses involving maternal, prenatal, and perinatal indicators [20, 21].

The data used in this study were accessed through the Platform for Data Science Applied to Health (PCDaS), an initiative by the Oswaldo Cruz Foundation (FIOCRUZ) that centralises public health datasets and offers access through user-friendly interfaces and APIs (available at: https://pcdas.icict.fiocruz.br/conjunto-de-dados/sistema-de-informacao-sobre-nascidos-vivos/dicionario-de-variaveis/; Data accessed: 28 March 2025). We included all live births from singleton and multiple pregnancies recorded across Brazil's 26 states and the Federal District. Additionally, we excluded births with missing or implausible gestational age data. The dataset is fully anonymized and publicly available, exempting the need for ethical review board approval in accordance with Brazilian Resolution 466/2012 and 674/2022, which governs research using public secondary data without personal identifiers.

Study variables

PTB was defined as any live birth before 37 completed weeks of gestation, following the World Health Organization's definition. The PTB prevalence rate was calculated by dividing the number of PTB by the total number of live births in the same location and year, then multiplying by 100. These rates were calculated for each Brazilian state and macro-region. PTB prevalence rates were calculated separately for each subgroup defined by maternal age, education level, race/ethnicity, prenatal care, and other relevant obstetric and neonatal factors.

The study variables were grouped into four main categories: sociodemographic, obstetric, prenatal, and neonatal characteristics. Sociodemographic variables included maternal age was categorized as < 20, 20–34, or ≥ 35 years; education level as no formal education, 1–3, 4–7, 8–11, or ≥ 12 years of schooling; marital status as single, married, or widowed; and race/ethnicity as White, Black, Brown (Pardo), Yellow (Asian), or Indigenous. Obstetric variables comprised type of pregnancy classified as singleton, twin, or triplet/multiple and mode of delivery vaginal or caesarean. Prenatal care was represented by the number of antenatal visits, categorized as 0, 1–3, 4–6, or ≥ 7. Neonatal variables included birthweight (< 2,500 g, 2,500–4,000 g, or > 4,000 g), sex (male or female), and the presence of congenital anomalies (yes or no). Only births with defined gestational age were included in the analysis.

Statistical analysis

Descriptive analyses were performed to summarise the distribution of PTB by maternal, obstetric, and neonatal characteristics. Categorical variables were presented as absolute frequencies and percentages n (%), facilitating the estimation of PTB prevalence within each subgroup. To examine subgroup differences, all analyses were stratified according to the key study variables: maternal age, education level, race/ethnicity, number of prenatal visits, and other relevant factors. Stratified analyses were conducted both descriptively and within the time-series framework, enabling comparison of PTB prevalence and temporal patterns across sociodemographic and obstetric subgroups.

Two complementary time-series techniques were applied: Prais–Winsten regression and SARIMA/GARCH models. The first estimated temporal trends and annual percentage changes accounting for autocorrelation, while the latter modelled forecasts and volatility.

To analyse temporal trends in PTB prevalence from 2014 to 2023, we employed the Prais Winsten linear regression model, which accounts for first-order serial autocorrelation typically present in time-series data. This method allows more accurate estimation of trends by adjusting for the dependence of each year's value on its preceding value. From this model, we calculated the Annual Percentage Change (APC) and its 95% confidence interval (95% CI) to quantify the yearly variation in PTB rates. The direction of trends was classified as increasing when the regression coefficient (β) was positive and statistically significant ($p < 0.05$), decreasing when β was negative and significant ($p < 0.05$), and stable when $p \geq 0.05$ [18, 22]. Additionally, the Average Annual Percent Change (AAPC) was computed to quantify the overall trend in PTB prevalence over the 10-year period for each Brazilian state. This approach is recommended

when annual data fluctuate and do not follow a strictly linear pattern, as described in the Joinpoint Regression Program developed by the U.S. National Cancer Institute [23].

The SARIMA model included the integrated differencing term (d), following standard notation. This combined approach enables robust estimation of historical trends and reliable projection of future rates, thereby enhancing the interpretability and analytical depth of the study. Spatial descriptive analysis was conducted to visualize geographic disparities in PTB prevalence across Brazil's 26 states and the Federal District, showing the spatial distribution of PTB rates in 2014 and 2023 and regional and temporal variations.

A national monthly time series of PTB rates was developed using SINASC data from January 2014 to December 2023. To identify temporal patterns, we applied Seasonal-Trend Decomposition using Loess (STL) and performed the Augmented Dickey–Fuller and KPSS tests to assess stationarity and trend components [24]. To forecast trends through 2030, we employed two modelling approaches: the Seasonal Autoregressive Integrated Moving Average (SARIMA) model and the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model [25–27]. The SARIMA model was fitted to capture both seasonal and non-seasonal components, with model selection guided by information criteria and residual diagnostics [26]. Model adequacy was assessed through several diagnostic tests, including the Ljung–Box test for residual autocorrelation, the Shapiro–Wilk test for normality, and the ARCH LM test for heteroskedasticity. When conditional heteroskedasticity was detected, the GARCH model was used to account for volatility clustering and irregular fluctuations in PTB rates over time [27]. Model performance errors (RMSE, MAE, and MAPE) were expressed in proportional units relative to PTB rates.

To assess social inequalities over time, we constructed stratified time-series models by maternal race/ethnicity and education level, applying SARIMA and GARCH analyses separately for each subgroup. R packages included *forecast* for ARIMA and SARIMA modelling, and *rugarch* for GARCH model estimation and forecasting with a 5% significance level. All analyses were conducted using R software (version 4.1.0; R Foundation for Statistical Computing, Vienna, Austria).

Results

Between 2014 and 2023, Brazil recorded over 25.5 million live births, of which 11.3% were preterm (Table 1). The prevalence of PTB varied considerably across maternal, prenatal, and neonatal characteristics, with a higher proportion of mothers aged ≥ 35 years and < 20 years presenting greater PTB prevalence (13.6% and 12.8%, respectively) compared with those where mothers aged 20–34 years predominated (10.5%). Regionally, PTB rates were higher in the North (12.2%) and Northeast (11.5%) compared with the South, Southeast, and Central-West regions. Across racial and ethnic groups, PTB was highest among Indigenous (15.0%) and Black (11.8%) populations. Complementary visualizations of annual and monthly PTB rates by state are available in the Figs. 1 and S1–S2.

Sociodemographic inequalities were evident, with larger proportions of women with low education (1–3 years of schooling) exhibited higher PTB prevalence (13.8%) than those where most mothers had ≥ 12 years (11.0%). Similarly, PTB proportions were higher in areas with limited prenatal care (23.2% with no visits vs. 8.4% with seven or more visits). Multiple pregnancies were strongly associated with PTB (93.5% among twins and 58.3% among triplets or more). PTB was also more frequent among male newborns (11.6%) and slightly higher in states with a predominance of caesarean deliveries (11.6%) than vaginal births (11.0%). At the national level, PTB prevalence increased modestly from 11.3% in 2014 to 11.9% in 2023. The annual percentage change (APC) was 0.72% (95% CI: 0.43 to 1.00; $p < 0.001$), while the average annual percentage change (AAPC) was 0.28%, indicating a statistically significant upward trend (Table 2). At the subnational level, 13 states exhibited significant increases, particularly Espírito Santo (AAPC: 2.11%), Roraima (2.62%), Sergipe (3.35%), Goiás (1.53%), Mato Grosso (1.66%), and São Paulo (0.63%). Only Maranhão showed a decreasing trend (AAPC: − 0.54%), while 13 states displayed stable rates throughout the period. The spatial distribution of PTB prevalence in 2014 and 2023 (Fig. 2) illustrates persistent regional disparities, with higher concentrations in northern and northeastern states. These patterns are further detailed in the Heatmap (Figure S2).

Temporal trend analysis confirmed regional and social heterogeneity (Table 3). States with greater proportions of older mothers, lower education, and higher representation of black, white, and yellow infants tended to show upward trajectories, whereas those with higher proportions of adolescents and Indigenous mothers exhibited stability. Additional disaggregated results are presented in Table S1.

Forecasting analyses using SARIMA and GARCH models (Figs. 3 and 4) projected a continued gradual increase in PTB rates through 2030. The SARIMA model captured long-term seasonality and short-term oscillations, while the GARCH model identified increasing conditional volatility, indicating greater forecast uncertainty rather than true fluctuations in PTB rates. Model diagnostics indicated satisfactory performance, with minimal autocorrelation (Ljung–Box $p = 0.588$) and strong predictive

Table 1 Descriptive statistics of maternal and neonatal characteristics according to preterm birth status in Brazil, 2014–2023

	Preterm Birth Status		
Variables	**Full-term**	**Preterm**	**Total**[1]
	N = 22,626,881 (88.7)[1]	*N* = 2,886,725 (11.3)[1]	*N* = 25,513,606
Mother's Age			
< 20 years	3,391,355 (87.2)	495,609 (12.8)	3,886,964
20–34 years	15,845,139 (89.5)	1,856,226 (10.5)	17,701,365
35 + years	3,390,387 (86.4)	534,890 (13.6)	3,925,277
Year			
2014	2,327,532 (88.7)	297,226 (11.3)	2,624,758
2015	2,373,663 (89.0)	292,273 (11.0)	2,665,936
2016	2,266,930 (88.8)	285,875 (11.2)	2,552,805
2017	2,340,027 (89.0)	289,929 (11.0)	2,629,956
2018	2,384,644 (88.9)	297,182 (11.1)	2,681,826
2019	2,320,993 (88.8)	291,808 (11.2)	2,612,801
2020	2,218,009 (88.6)	284,828 (11.4)	2,502,837
2021	2,178,574 (88.6)	279,092 (11.4)	2,457,666
2022	2,106,026 (88.2)	282,182 (11.8)	2,388,208
2023	2,110,483 (88.1)	286,330 (11.9)	2,396,813
Marital Status			
Consensual Union	4,598,506 (88.7)	587,479 (11.3)	5,185,985
Divorced	302,036 (87.6)	42,746 (12.4)	344,782
Married	7,475,294 (89.2)	907,631 (10.8)	8,382,925
Single	10,212,513 (88.4)	1,343,399 (11.6)	11,555,912
Widowed	38,532 (87.6)	5,470 (12.4)	44,002
Region			
Central-West	1,827,851 (88.9)	227,924 (11.1)	2,055,775
North	2,393,534 (87.8)	332,907 (12.2)	2,726,441
Northeast	5,981,409 (88.6)	766,232 (11.4)	6,747,641
South	3,280,319 (88.9)	408,597 (11.1)	3,688,916
Southeast	9,143,768 (88.8)	1,151,065 (11.2)	10,294,833
Years of Schooling			
None	83,048 (84.9)	14,782 (15.1)	97,830
1–3 years	391,435 (86.2)	62,651 (13.8)	454,086
4–7 years	3,260,591 (87.4)	471,168 (12.6)	3,731,759
8–11 years	14,097,484 (89.0)	1,742,615 (11.0)	15,840,099
12 + years	4,794,323 (89.0)	595,509 (11.0)	5,389,832
Type of Pregnancy			
Single fetus	22,401,678 (89.8)	2,556,804 (10.2)	24,958,482
Twins	224,393 (41.3)	318,346 (58.7)	542,739
Triplets+	810 (6.5)	11,575 (93.5)	12,385
Birth Weight			
< 2500 g	859,579 (39.1)	1,340,535 (60.9)	2,200,114
2500–4000 g	20,594,544 (93.2)	1,509,156 (6.8)	22,103,700
>4000 g	1,172,758 (96.9)	37,034 (3.1)	1,209,792
Sex of Newborn			
Female	11,088,738 (89.1)	1,359,088 (10.9)	12,447,826
Male	11,538,143 (88.3)	1,527,637 (11.7)	13,065,780
Congenital Anomaly	169,659 (75.6)	54,718 (24.4)	224,377
Maternal Race/Ethnicity			
Black	1,415,279 (88.2)	189,189 (11.8)	1,604,468
Brown	12,709,562 (88.7)	1,618,351 (11.3)	14,327,913
Indigenous	189,201 (85.0)	33,410 (15.0)	222,611
White	8,213,829 (88.8)	1,032,999 (11.2)	9,246,828
Yellow	99,010 (88.6)	12,776 (11.4)	111,786

Table 1 (continued)

	Preterm Birth Status		
Type of Delivery			
Caesarean	12,902,958 (88.4)	1,687,534 (11.6)	14,590,492
Vaginal	9,723,923 (89.0)	1,199,191 (11.0)	10,923,114
Prenatal Visits			
0	114,337 (76.8)	34,593 (23.2)	148,930
1–3	1,114,469 (77.2)	329,751 (22.8)	1,444,220
4–6	4,486,221 (82.1)	979,019 (17.9)	5,465,240
7+	16,911,854 (91.6)	1,543,362 (8.4)	18,455,216

[1]n (%)

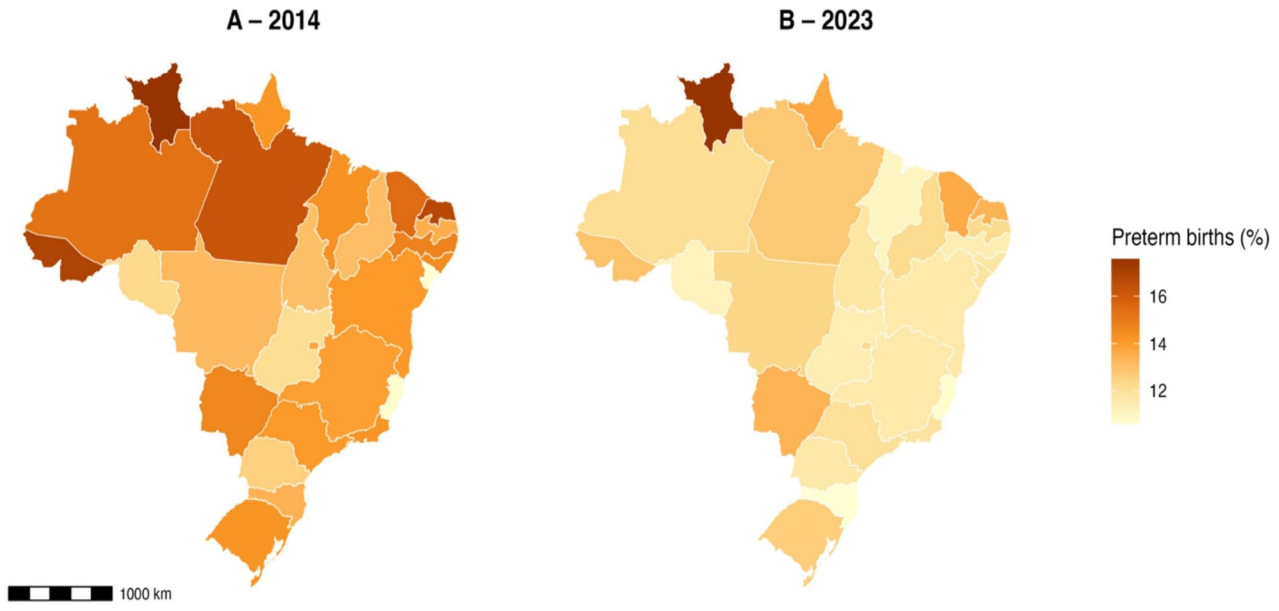

Fig. 1 Spatial distribution of preterm birth prevalence across Brazilian states in 2014 (**A**) and 2023 (**B**)

accuracy (RMSE = 0.0020; MAPE = 1.33%). Seasonal fluctuations and monthly variations are shown in Figure S4.

Stratified forecasts by maternal characteristics (Figs. 5 and 6) revealed persistent social inequalities. States with larger proportions of Indigenous and Black mothers are projected to exceed 15% PTB prevalence by 2030, whereas those with predominantly White and Yellow populations are expected to remain below 13%. Similarly, forecasts by maternal education level with higher proportions of women with no formal schooling may reach or surpass 30% PTB prevalence, while those where most mothers have ≥ 12 years of education are projected to maintain lower and stable rates. These subgroup projections are illustrated in Figure S3.

Discussion

Between 2014 and 2023, Brazil recorded over 25.5 million live births, with an average PTB prevalence of 11.3%, equivalent to approximately 1 in every 9 births. In absolute numbers, around 2.8 million PTB were documented, showing an increasing trend over the period. This prevalence was especially high among adolescent mothers (< 20 years), women aged 35 or older, those with low educational attainment, without prenatal care, and among Black and Indigenous women. Although the national prevalence remained relatively stable, a statistically significant upward trend was observed, with a notable increase in 13 states, particularly in the Central-West region, and a reduction only in Maranhão. The SARIMA and GARCH models applied to the monthly time series confirmed this upward trend, identified seasonal patterns, and revealed increasing uncertainty over time. Projections indicate a continuous rise in PTB rates through 2030, with growing disparities among Indigenous, Black, and low-educated mothers.

These findings align with the national literature [8, 16]. An ecological study using SINASC data found that the prevalence of PTB in Brazil between 2011 and 2021 was 11.1%. The prevalence remained stable overall, with higher in the North region, and increases among twin pregnancies and mothers who had 4 to 6 prenatal visits [16]. A cohort study of Brazilian women between 2012

Table 2 Trends in preterm birth prevalence by state in Brazil (2014–2023): APC, AAPC, and trend classification

State	Prevalence		APC	AAPC	CI 95	*P*- value	Trend
	2014	2023					
Acre	13.4	12.9	-0.55	-0.41	(-1.82; 0.73)	0.422	Stable
Alagoas	11.7	11.7	-0.05	0.09	(-1.16; 1.07)	0.930	Stable
Amapá	11.4	13.7	0.70	2.02	(-3.92; 5.55)	0.778	Stable
Amazonas	12.2	12.1	0.57	-0.06	(-0.46; 1.62)	0.310	Stable
Bahia	11.3	11.6	0.07	0.21	(-0.47; 0.61)	0.803	Stable
Ceará	12.4	13.6	1.45	1.03	(0.69; 2.23)	0.006	Increasing
Distrito Federal	11.1	12.2	0.07	1.08	(-0.99; 1.15)	0.895	Stable
Espírito Santo	8.9	10.7	1.86	2.11	(1.31; 2.41)	0.000	Increasing
Goiás	9.9	11.4	1.85	1.53	(0.97; 2.75)	0.003	Increasing
Maranhão	11.5	11.0	-0.91	-0.54	(-1.63; -0.19)	0.038	Decreasing
Mato Grosso	10.7	12.4	1.94	1.66	(1.03; 2.85)	0.003	Increasing
Mato Grosso do Sul	11.8	13.3	1.41	1.40	(0.64; 2.17)	0.007	Increasing
Minas Gerais	11.2	11.6	0.23	0.31	(-0.09; 0.56)	0.196	Stable
Paraná	10.3	11.6	1.63	1.37	(1.15; 2.11)	0.000	Increasing
Paraíba	10.9	12.3	1.42	1.28	(0.65; 2.21)	0.007	Increasing
Pará	12.9	12.8	0.00	-0.13	(-0.3; 0.3)	0.990	Stable
Pernambuco	11.9	11.4	-0.29	-0.46	(-0.98; 0.41)	0.438	Stable
Piauí	10.6	12.2	1.37	1.59	(0.18; 2.58)	0.054	Stable
Rio Grande do Norte	13.3	13.3	0.05	0.06	(-0.55; 0.65)	0.877	Stable
Rio Grande do Sul	11.5	12.6	1.31	1.10	(0.95; 1.68)	0.000	Increasing
Rio de Janeiro	11.4	11.9	0.61	0.40	(-0.13; 1.37)	0.147	Stable
Rondônia	10.1	11.1	1.58	1.14	(0.56; 2.62)	0.016	Increasing
Roraima	13.9	17.6	2.39	2.62	(0.67; 4.14)	0.026	Increasing
Santa Catarina	10.9	10.6	-0.25	-0.28	(-0.71; 0.22)	0.331	Stable
Sergipe	8.8	11.8	1.88	3.35	(0.61; 3.17)	0.020	Increasing
São Paulo	11.3	12.0	1.00	0.63	(0.53; 1.47)	0.003	Increasing
Tocantins	10.6	11.8	1.21	1.16	(0.64; 1.78)	0.003	Increasing
Brazil	**11.3**	**11.9**	**0.72**	**0.28**	**(0.43; 1.00)**	**0.001**	**Increasing**

Abbreviations: APC, annual percentage change; AAPC, average annual percentage change; 95 CI, 95 confidence interval

and 2019 highlighted ethnic-racial disparities in neonatal outcomes, with Indigenous and Black women experiencing higher proportions of PTB, low birth weight, and early neonatal mortality [28]. Similar results have been observed in the United States, where PTB was more prevalent among adolescents, older mothers, vulnerable populations, and residents in socioeconomically disadvantaged areas [11]. These findings are consistent with the classic study by Goldenberg and colleagues, who identified higher prevalence of PTB among Black mothers with low education, low income, and multiple pregnancies [14].The literature reinforces that PTB is strongly associated with social and biomedical determinants, including infections, chronic stress, exposure to adverse environments, and structural inequalities [2, 3, 14, 29].

PTB in Brazil has been characterised as a perinatal "epidemic" [30], driven by multiple socioeconomic and clinical factors, especially among low-income and low-educated women. Conditions such as infections during pregnancy, gestational diabetes, smoking, and history of PTB are well-documented risk factors [14, 31]. Moreover, caesarean deliveries have been strongly associated with PTB, often performed for medical or maternal convenience, without clinical indication [32–35]. It is estimated that approximately 26% of PTB in Brazil are related to caesarean sections performed before labour onset [30]. The findings of this study, indicating an increasing trend of PTB with marked regional variations, reinforce the hypothesis that Brazil faces a silent epidemic, strongly influenced by high rates of elective caesarean deliveries without clinical indication particularly among women with higher education, used here as a proxy for socioeconomic status, and older maternal age [30].

Regarding ethnicity, although trends remained stable among Black and Indigenous women, a significant increase in PTB prevalence was observed among White, mixed-race (Pardo), and yellow newborns. The literature consistently shows an association between ethnicity and PTB, with Black women historically presenting higher rates [36]. However, recent studies suggest that all racialized or minoritized groups tend to show higher rates of PTB compared to White women [37]. During the pandemic, these disparities were further accentuated: Black women were at higher risk of adverse outcomes,

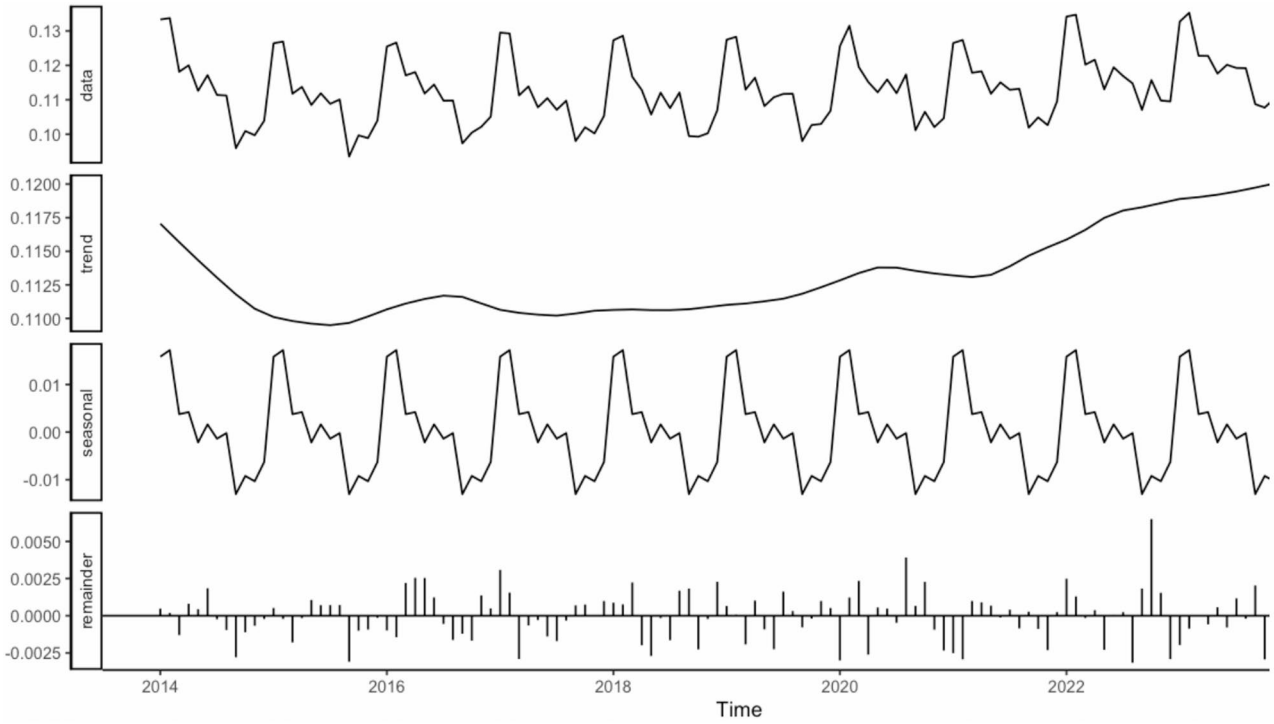

Fig. 2 STL decomposition of monthly preterm birth rates in Brazil from 2014 to 2023, showing the observed data, long-term trend, seasonal component, and remainder (residuals)

Table 3 Annual trends in preterm birth according to region, maternal education, race/ethnicity, and age group in Brazil, 2014–2023

Variables	Prevalence		APC	AAPC	CI 95	*P*- value	Trend
	2014	2023					
Region	10.7	12.1	1.44	1.41	(0.95; 1.92)	0.000	Increasing
Central-West	10.7	12.1	1.44	1.41	(0.95; 1.92)	0.000	Increasing
North	12.3	12.7	0.49	0.32	(0.11; 0.87)	0.037	Increasing
Northeast	11.6	12.0	0.32	0.38	(-0.31; 0.96)	0.343	Stable
South	10.8	11.7	1.01	0.84	(0.69; 1.34)	0.000	Increasing
Southeast	11.2	11.8	0.78	0.56	(0.36; 1.21)	0.007	Increasing
Education							
1–3 years	13.6	15.5	1.40	1.49	(0.83; 1.98)	0.001	Increasing
12 + years	11.0	11.4	0.61	0.42	(0.32; 0.89)	0.003	Increasing
4–7 years	12.6	13.9	1.13	1.08	(0.54; 1.73)	0.006	Increasing
8–11 years	10.9	11.8	1.01	0.92	(0.58; 1.44)	0.002	Increasing
None	15.2	17.7	2.16	1.69	(0.53; 3.83)	0.032	Increasing
Newborn Race/Ethnicity							
Black	12.1	12.2	0.22	0.10	(-0.29; 0.74)	0.423	Stable
Brown	11.4	12.0	0.68	0.60	(0.19; 1.17)	0.027	Increasing
Indigenous	16.1	16.2	0.13	0.07	(-1.19; 1.47)	0.852	Stable
White	11.1	11.7	0.78	0.59	(0.47; 1.11)	0.001	Increasing
Yellow	11.5	11.9	0.64	0.40	(0.11; 1.16)	0.044	Increasing
Maternal Age							
20–34 years	10.5	11.1	0.70	0.59	(0.26; 1.15)	0.014	Increasing
35 + years	13.2	14.6	1.27	1.09	(0.94; 1.61)	0.000	Increasing
< 20 years	13.1	13.1	0.08	0.02	(-0.42; 0.58)	0.762	Stable

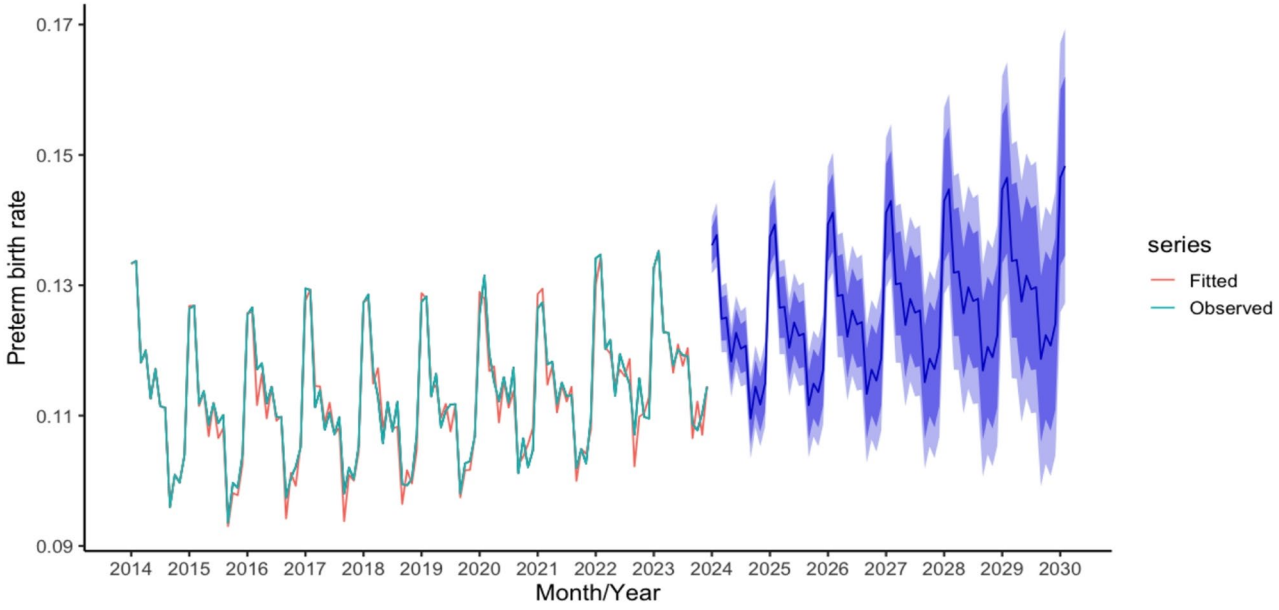

Fig. 3 SARIMA-based monthly forecast of preterm birth rates in Brazil from December 2023 to December 2030, with 80% and 95% prediction intervals

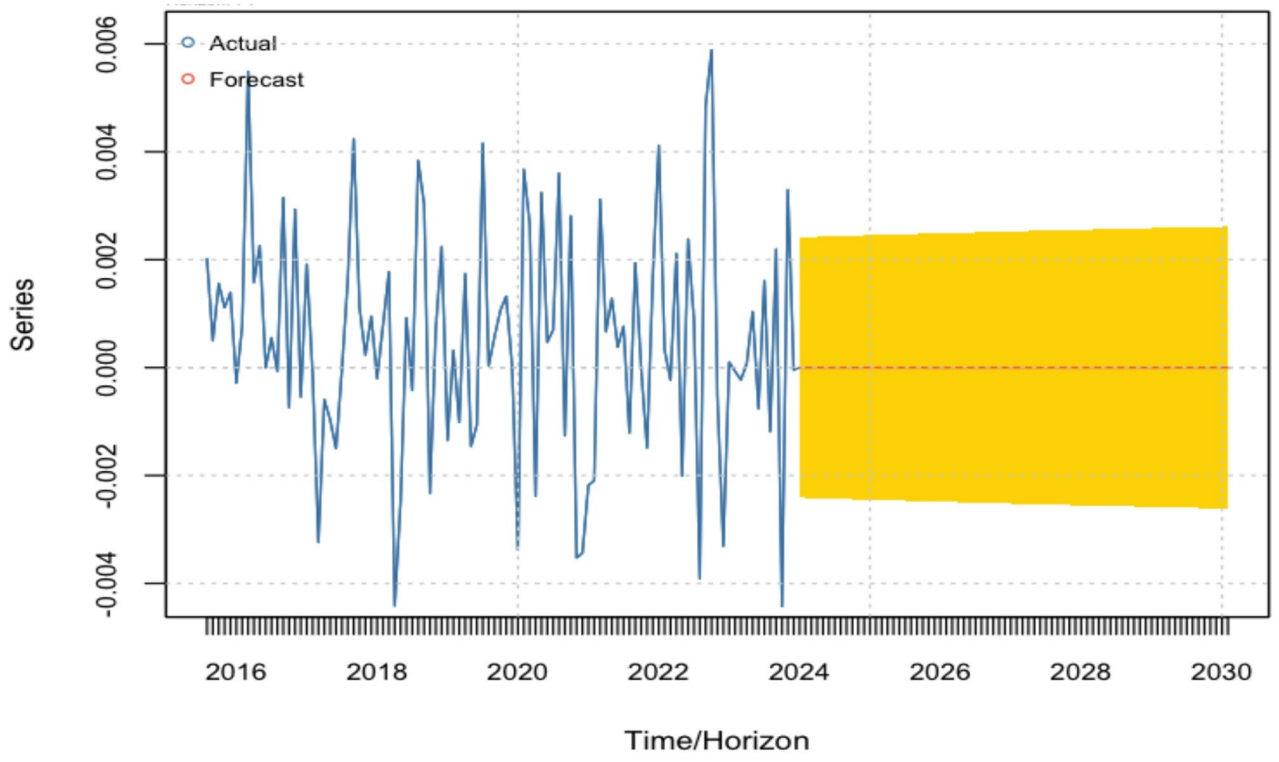

Fig. 4 GARCH forecast of preterm birth rate residuals with 1-sigma confidence bands from 2024 to 2030, showing conditional variance estimates over time

including higher rates of ICU admission, severe acute respiratory syndrome, and maternal mortality [38]. Racial and socioeconomic disparities persist, with babies born to Black mothers being 2.5 times more likely to be born very preterm compared to those born to White mothers [3].

These racial and socioeconomic inequalities remain key determinants. A national study conducted in England with over 1 million births indicated that 18.5% of PTB could be attributed to socioeconomic inequality and 1.2% to ethnic inequality [39]. Black and Indigenous women face higher risks of adverse birth outcomes, including

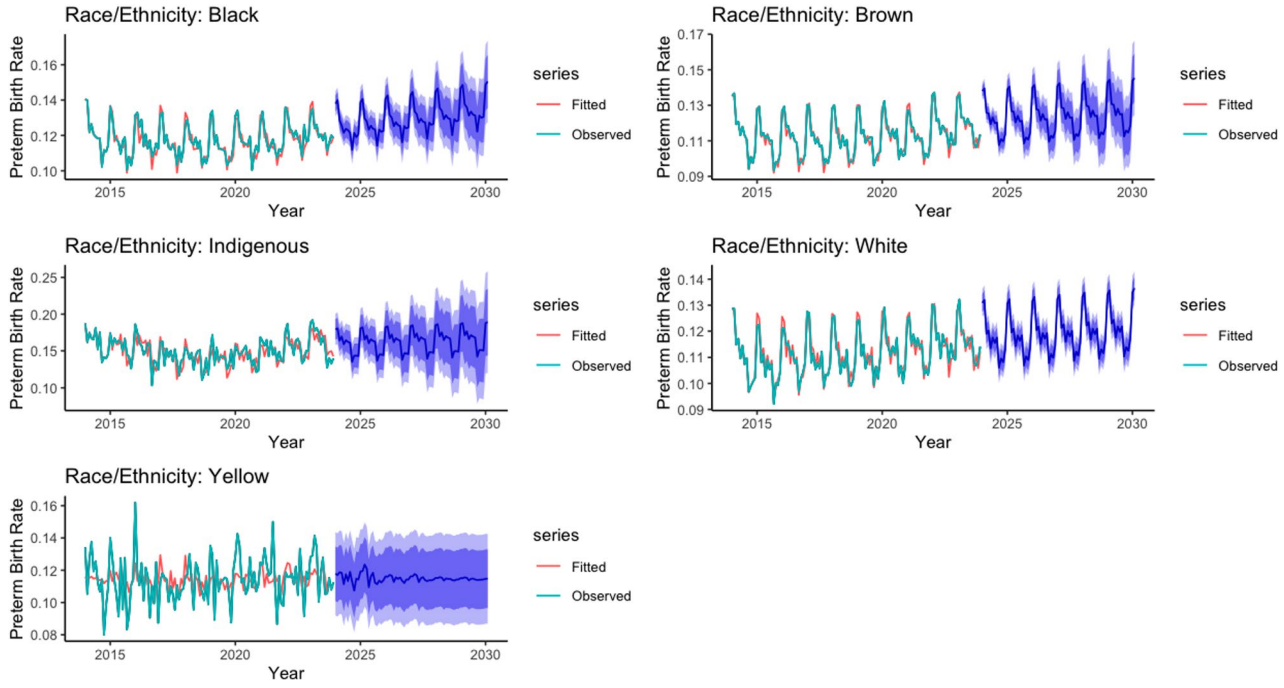

Fig. 5 Forecasts of preterm birth rates by maternal race/ethnicity in Brazil from 2014 to 2030 using SARIMA models. Solid lines represent observed values (2014–2023) and fitted/forecasted trends; shaded areas denote 80% and 95% confidence intervals

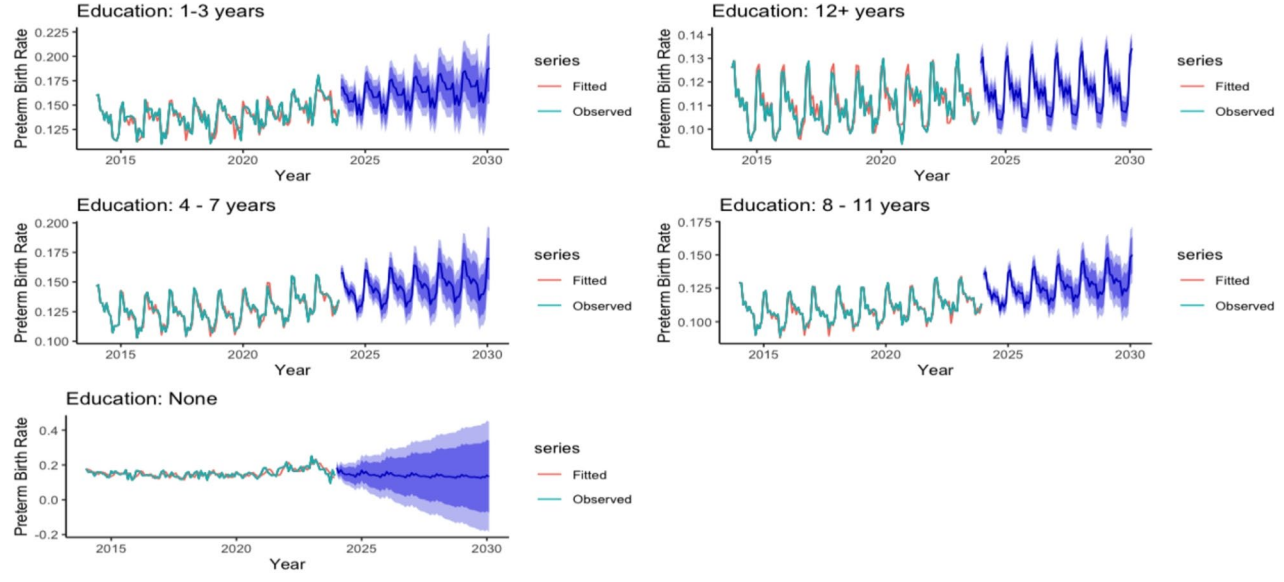

Fig. 6 Forecasted preterm birth rates in Brazil by maternal education level using SARIMA models (2014–2030) and fitted/forecasted trends; shaded areas denote 80% and 95% confidence intervals

PTB, largely due to mechanisms of structural racism that affect access to and quality of available health services [28]. A study using Bayesian spatial models to quantify the risks of stillbirths and PTB in Philadelphia found associations with neighbourhood characteristics, such as poverty and violence rates, reinforcing the influence of contextual social determinants [40].

Regarding maternal age, a stable trend was observed among adolescents (< 20 years) a group that, despite having one of the highest prevalences, did not show significant variation over the period. On the other hand, women aged 35 years or older showed an increase in PTB rates. This finding is consistent with previous studies [41, 42], which demonstrate higher risk of PTB among older mothers. Although both age groups share similar risk

factors such as low socioeconomic status, extreme BMI, and smoking older women tend to have a higher prevalence of comorbidities like hypertension and gestational diabetes, which may contribute to increasing rates [43, 44].

Projections through 2030 indicate a gradual upward trend in PTB rates, both nationally and among vulnerable subgroups such as Indigenous, Black, and uneducated mothers. This demonstrates that PTB remains a pressing and growing public health challenge in Brazil, with a strong component of social inequality. Public policies aimed at expanding access to and improving the quality of prenatal care, addressing structural inequalities, and continuously monitoring maternal and child health indicators are essential strategies to mitigate the observed trends [45, 46].

This study presents several methodological and analytical strengths, particularly the use of SINASC data a comprehensive national database with national coverage, recognized for its completeness, consistency, and breadth of sociodemographic and obstetric variables [47, 48]. The large sample size and long observation period (2014–2023) allowed robust analyses of temporal and regional patterns, highlighting seasonal fluctuations and the impact of events such as the COVID-19 pandemic. The application of advanced time-series models (SARIMA and GARCH), combined with diagnostic and heteroskedasticity tests, improved the precision and robustness of the forecasts. Stratification by race/ethnicity and maternal education also provided valuable insights into persistent social inequalities in preterm birth in Brazil. Nevertheless, several limitations should be considered. PTB rates were presented as crude proportions of live births, which may not fully account for regional differences in maternal characteristics such as age or education. Still, this approach was appropriate for describing temporal trends and forecasts. Future research should include standardized or adjusted rates to improve regional comparability. Second, despite the high quality of SINASC data, some variables, especially gestational age and race/ethnicity, may be incomplete or inconsistently recorded in certain municipalities, which could lead to misclassification or measurement bias. Third, the database lacks detailed clinical information such as maternal comorbidities, infections, and medication use, which could help explain biological mechanisms underlying PTB trends. Fourth, as an ecological study, the associations identified cannot be interpreted as causal, and residual confounding by unmeasured factors may remain. Finally, forecasting models are inherently dependent on the quality of historical data and the assumption that past patterns will continue in the future, which introduces uncertainty into long-term projections.

Conclusion

This study demonstrates an upward trend in preterm birth rates in Brazil over the past decade, with persistent regional and social inequalities, particularly among Indigenous, Black, and low-educated mothers. Time-series forecasting using SARIMA and GARCH models projects a continued increase through 2030 if no preventive actions are taken. These findings underscore the need for targeted, equity-oriented public health interventions such as expanding access to quality prenatal care and improving maternal education to reduce preventable PTB and promote more equitable maternal and neonatal outcomes nationwide.

Data Availability

The data analyzed in this study are publicly available through SINASC, which is managed by the Brazilian Ministry of Health. These data were accessed via the Platform for Data Science Applied to Health (PCDaS), an initiative of the Oswaldo Cruz Foundation (FIOCRUZ) that centralizes public health datasets and provides access through user-friendly interfaces and APIs (available at: https://pcdas.icict.fiocruz.br/conjunto-de-dados/sistema-de-informacao-sobre-nascidos-vivos/dicionario-de-variaveis/).

Ethics statement

Ethical approval was not required for this study, as it utilized anonymized, publicly accessible data without personal identifiers, in compliance with Resolution 466/2012 of the Brazilian National Health Council, which governs ethics in research involving human subjects.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12884-025-08536-6.

Supplementary Material 1

Acknowledgements
Not applicable.

Author contributions
AV, TCS, and PHCR conceptualized and designed the study. AV, SPX, HAFB, VAM, ARE, and CMC developed the methodology. AV, TCS, PHCR, HAFB, VAM, SX, and AMN carried out the investigation. Data curation, analysis, and visualization were performed by AV and AMN. AV and HAFB wrote the original draft. PHCR, TCS, and CMC provided supervision and critical guidance. Writing review and editing were conducted by AV, SPX, HAFB, VAM, ARE, and CMC. All authors contributed to the manuscript, reviewed and approved the final version, and are accountable for the content and integrity of the work.

Funding
A.V. received scholarships from FAPESP (grant number 2023/07936-3) and FAPESP-BEPE (grant number 2024/18309-2).

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Declarations

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Conflict of interest
The authors declare no conflicts of interest.

Word count
3402.

Conflict of interest
The authors declare that they have no conflicts of interest.

Consent to participate declaration:
Not applicable.

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Received: 17 June 2025 / Accepted: 25 November 2025
Published online: 05 December 2025

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